12.1 MIXED KALMAN/H_∞ FILTERING

In this section we look at the problem of finding a filter that combines the best features of Kalman filtering with the best features of H_∞ filtering. This problem can be attacked a couple of different ways. Recall from Section 5.2 the cost function that is minimized by the steady-state Kalman filter:

$$J_2 = \lim_{N\to\infty} \sum_{k=0}^{N} E\left(||x_k - \hat{x}_k||_2\right) \tag{12.1}$$

Recall from Section 11.3 the cost function that is minimized by the steady-state H_∞ state estimator if S_k and L_k are identity matrices:

$$J_\infty = \lim_{N\to\infty} \max_{x_0, w_k, v_k} \frac{\sum_{k=0}^{N} ||x_k - \hat{x}_k||^2}{||x(0) - \hat{x}(0)||^2_{P_0^{-1}} + \sum_{k=0}^{N}\left(||w_k||^2_{Q_k^{-1}} + ||v_k||^2_{R_k^{-1}}\right)} \tag{12.2}$$

Loosely speaking, the Kalman filter minimizes the RMS estimation error, and the H_∞ filter minimizes the worst-case estimation error.

In [Had91] these two performance objectives are combined to form the following problem: Given the n-state observable LTI system

$$\begin{aligned} x_{k+1} &= Fx_k + w_k \\ y_k &= Hx_k + v_k \end{aligned} \tag{12.3}$$

where $\{w_k\}$ and $\{v_k\}$ are uncorrelated zero-mean, white noise processes with covariances Q and R respectively, find an estimator of the form

$$\hat{x}_{k+1} = \hat{F}x_k + Ky_k \tag{12.4}$$

that satisfies the following criteria:

1. $\hat{F}$ is a stable matrix (so the estimator is stable).
2. The H_∞ cost function is bounded by a user-specified parameter:

$$J_\infty < \frac{1}{\theta} \tag{12.5}$$

3. Among all estimators satisfying the above criteria, the filter minimizes the Kalman filter cost function J_2.

The solution to this problem provides the best RMS estimation error among all estimators that bound the worst-case estimation error. The filter that solves this problem is given as follows.

The mixed Kalman/H_∞ filter

1. Find the $n \times n$ positive semidefinite matrix P that satisfies the following Riccati equation:

$$P = FPF^T + Q + FP(I/\theta^2 - P)^{-1}PF^T - P_aV^{-1}P_a^T \tag{12.6}$$

where P_a and V are defined as

$$\begin{aligned} P_a &= FPH^T + FP(I/\theta^2 - P)^{-1}PH^T \\ V &= R + HPH^T + HP(I/\theta^2 - P)^{-1}PH^T \end{aligned} \tag{12.7}$$

2. Derive the $\hat{F}$ and K matrices in Equation (12.4) as

$$\begin{aligned} K &= P_a V^{-1} \\ \hat{F} &= F - KH \end{aligned} \tag{12.8}$$

3. The estimator of Equation (12.4) satisfies the mixed Kalman/H$_\infty$ estimation problem if and only if $\hat{F}$ is stable. In this case, the state estimation error satisfies the bound

$$\lim_{k\to\infty} E\left(||x_k - \hat{x}_k||^2\right) \le \mathrm{Tr}(P) \tag{12.9}$$

Note that if $\theta = 0$, then the problem statement reduces to the Kalman filter problem statement. In this case we can see that Equation (12.6) reduces to the discrete-time algebraic Riccati equation that is associated with the Kalman filter (see Problem 12.2 and Section 7.3). The continuous-time version of this theory is given in [Ber89].

■ EXAMPLE 12.1

In this example, we take another look at the scalar system that is described in Example 11.2:

$$\begin{aligned} x_{k+1} &= x_k + w_k \\ y_k &= x_k + v_k \end{aligned} \tag{12.10}$$

where $\{w_k\}$ and $\{v_k\}$ are uncorrelated zero-mean, white noise processes with covariances Q and R, respectively. Equation (12.6), the Riccati equation for the mixed Kalman/H$_\infty$ filter, reduces to the following scalar equation:

$$\theta^2(1 - R\theta^2)P^3 + (Q\theta^2 + 1)(R\theta^2 - 1)P^2 + Q(1 - 2R\theta^2) + QR = 0 \tag{12.11}$$

Suppose that (for some value of Q, R, and θ) this equation has a solution $P \ge 0$, and $|1 - K| < 1$, where the filter gain K from Equation (12.8) is given as

$$K = \frac{P}{P + R - PR\theta^2} \tag{12.12}$$

Then J_∞ from Equation (12.2) is bounded from above by $1/\theta$, and the variance of the state estimation error is bounded from above by P. The top half of Figure 12.1 shows the Kalman filter performance bound P and the estimator gain K as a function of θ when $Q = R = 1$. Note that at $\theta = 0$ the mixed Kalman/H$_\infty$ filter reduces to a standard Kalman filter. In this case the performance bound $P \approx 1.62$ and the estimator gain $K \approx 0.62$, as discussed in Example 11.2. However, if $\theta = 0$ then we do not have any guarantee on the worst-case performance index J_∞.

From the top half of Figure 12.1, we see that as θ increases, the performance bound P increases, which means that our Kalman performance index gets worse. However, at the same time, the worst-case performance index J_∞ decreases as θ increases. From the bottom half of Figure 12.1, we see that as θ increases, K increases, which is consistent with better H_∞ performance and worse Kalman performance (see Example 11.2). When θ reaches about 0.91, numerical difficulties prevent a solution to the mixed filter problem.

The bottom half of Figure 12.1 shows that at $\theta = 0.5$ the estimator gain $K \approx 0.76$. Recall from Example 11.2 that the H_∞ filter had an estimator gain $K = 1$ for the same value of θ. This shows that the mixed Kalman/H_∞ filter has a smaller estimator gain (for the same θ) than the pure H_∞ filter. In other words, the mixed filter uses a lower gain in order to obtain better Kalman performance, whereas the pure H_∞ filter uses a higher gain because it does not take Kalman performance into account.

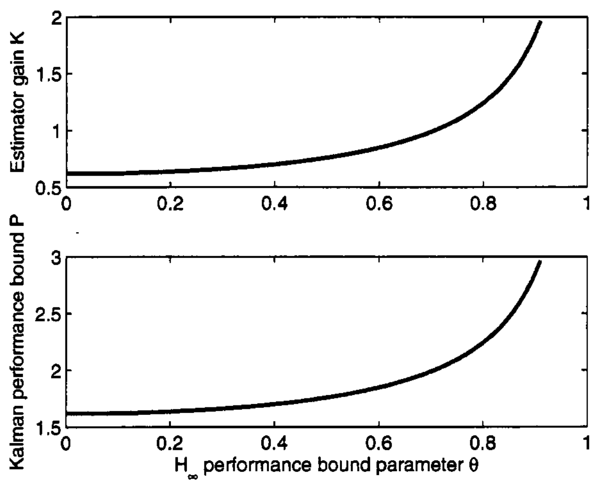

Figure 12.1 Results for Example 12.1 of the estimation-error variance bound and estimator gain as a function of θ for the mixed Kalman/H_∞ filter. As θ increases, the worst-case performance bound $1/\theta$ decreases, the error-variance bound increases, and the estimator gain increases greater than the Kalman gain ($\theta = 0$). This shows a trade-off between worst-case performance and RMS performance.

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Although the approach presented above is a theoretically elegant method of obtaining a mixed Kalman/H_∞ filter, the solution of the Riccati equation can be challenging for problems with a large number of states. Other more straightforward approaches can be used to combine the Kalman and H_∞ filters. For example, if the steady-state Kalman filter gain for a given problem is denoted as K_2 and the steady-state H_∞ filter gain is denoted as K_∞, then a hybrid filter gain can be constructed as

$$K = dK_2 + (1-d)K_\infty \tag{12.13}$$

where $d \in [0,1]$. This hybrid filter gain is a convex combination of the Kalman and H_∞ filter gains, which would be expected to provide a balance between RMS and worst-case performance [Sim96]. However, this approach is not as attractive theoretically since stability must be determined numerically, and no *a priori* bounds on